

STA 332/MATH 343: Homework 3

Duke University, Fall 2026

Due: September 25th, 11:59 PM on Gradescope

Constructing Estimators

1. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, \beta)$, where both the shape $\alpha > 0$ and the rate $\beta > 0$ are unknown. Find the method-of-moments estimators of α and β using the first two moments. Write your answer in terms of

$$m_1 = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad m_2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

2. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$.

- (a) Suppose σ^2 is known. Find the MLE of μ .
- (b) Suppose μ is known. Find the MLE of σ^2 .

3. In a simple model of inheritance, a person's genotype is AA, aa, or Aa. The two alleles are inherited independently and have the same distribution, with

$$\mathbb{P}_\theta(\text{A}) = \theta, \quad \mathbb{P}_\theta(\text{a}) = 1 - \theta,$$

where $0 < \theta < 1$. The order of the alleles does not matter, so Aa and aA represent the same genotype.

- (a) Compute $\mathbb{P}_\theta(\text{Aa})$.
- (b) Encode the genotype as

$$X = \begin{cases} (1, 0, 0), & \text{if the genotype is AA,} \\ (0, 1, 0), & \text{if the genotype is aa,} \\ (0, 0, 1), & \text{if the genotype is Aa.} \end{cases}$$

Find the pmf $p_\theta(x)$ for $x \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

- (c) A sample of 942 males from Ireland contains 501 of genotype AA, 83 of genotype aa, and 358 of genotype Aa. Find the MLE of θ .
4. The lifetime X of a cell-phone battery has density

$$f_\theta(x) = K\theta^x \mathbb{1}_{\{x>0\}},$$

where $0 < \theta < 1$ is unknown and K is the normalizing constant. Let $X_1, \dots, X_n$ be independent observations from this model.

- (a) Show that $K = \log(1/\theta)$.
 - (b) Rewrite the density in terms of $\eta = \log(1/\theta)$, and identify the resulting distribution of X (it is a named family). Write the statistical model in terms of η .
 - (c) Find the MLE $\hat{\eta}$, then use $\theta = e^{-\eta}$ and invariance of the MLE to find $\hat{\theta}$.
5. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\theta, \theta + 1)$, where $\theta \in \mathbb{R}$. Find the set of all maximum likelihood estimators of θ (there won't be just one MLE!).

Evaluating estimators

6. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, \theta)$, where $\theta > 0$. Consider the method-of-moments estimator and the maximum likelihood estimator

$$\hat{\theta}_{\text{MoM}} = 2\bar{X}_n, \quad \hat{\theta}_{\text{MLE}} = X_{(n)} = \max(X_1, \dots, X_n).$$

- (a) Compute the bias, variance, and MSE of $\hat{\theta}_{\text{MoM}}$.
 - (b) Compute the cdf and then the pdf of $\hat{\theta}_{\text{MLE}}$.
 - (c) Compute the bias, variance, and MSE of $\hat{\theta}_{\text{MLE}}$.
 - (d) Determine for which n the MSE of $\hat{\theta}_{\text{MLE}}$ is smaller than that of $\hat{\theta}_{\text{MoM}}$ for every $\theta > 0$.
7. Let $X_1, \dots, X_n$ be i.i.d. random variables with density

$$f_\lambda(x) = \frac{1}{\lambda} e^{-x/\lambda} \mathbb{1}_{\{x \geq 0\}},$$

where $\lambda > 0$ is unknown.

- (a) Find the distribution of $X_{(1)} = \min_i X_i$ (Hint: use the trick from HW1, Problem 10).
- (b) Use the previous part to give an unbiased estimator of λ .
- (c) Compute the MSE of the estimator from part (b).
- (d) Compute the MSE of $\bar{X}_n$. Which of the two estimators has smaller MSE when $n > 1$?